

Production Deployment Guide

Step-by-step setup for the Maria Glass & Plywood website on a fresh AWS EC2 instance running Ubuntu 26.04 LTS. Stack: React (built & served by Nginx) + FastAPI (uvicorn + systemd) + MongoDB 7 + Nginx reverse proxy + Let's Encrypt SSL.

0. Architecture Overview

This guide deploys the application in a single-server topology suitable for small to medium traffic:

- **Nginx** on ports 80/443 — serves the built React static files and reverse-proxies `/api/*` to the backend.
- **FastAPI (uvicorn)** on `127.0.0.1:8001` — managed by **systemd**.
- **MongoDB 7** on `127.0.0.1:27017` — local instance, bound to loopback.
- **Let's Encrypt** via certbot — auto-renewing HTTPS certificate.
- **UFW firewall** — only ports 22, 80, 443 are exposed externally.

1. Launch the EC2 Instance

1.1 Recommended instance

- AMI: **Ubuntu Server 26.04 LTS** (64-bit, x86_64)
- Instance type: **t3.small** minimum (2 vCPU, 2 GB RAM). **t3.medium** recommended.
- Storage: **30 GB gp3** EBS volume.
- Key pair: create or reuse an existing `.pem` key.

1.2 Security Group (inbound rules)

Type	Protocol	Port range	Source	Description
SSH	TCP	22	Your IP/32	Admin access
HTTP	TCP	80	0.0.0.0/0	Public web
HTTPS	TCP	443	0.0.0.0/0	Public web (SSL)

1.3 Elastic IP & DNS

Allocate an Elastic IP and associate it to the instance. Then in your DNS provider (e.g. GoDaddy, Cloudflare, Route 53), create an A-record:

<code>mariaglassplywood.com</code>	A	<code><ELASTIC_IP></code>
<code>www.mariaglassplywood.com</code>	A	<code><ELASTIC_IP></code>

Wait until `dig mariaglassplywood.com` resolves to the Elastic IP before requesting SSL certificates.

2. Connect & Update the Server

```
chmod 400 ~/maria-key.pem
ssh -i ~/maria-key.pem ubuntu@<ELASTIC_IP>
```

```
# Update system
sudo apt update && sudo apt upgrade -y
sudo timedatectl set-timezone Asia/Kolkata
```

Create a deploy user (optional but recommended)

```
sudo adduser deploy --gecos "" --disabled-password
sudo usermod -aG sudo deploy
sudo rsync --archive --chown=deploy:deploy ~/.ssh /home/deploy
```

```
# From now on: ssh -i ~/maria-key.pem deploy@<ELASTIC_IP>
```

3. Install Core Dependencies

3.1 System packages

```
sudo apt install -y build-essential curl git unzip ca-certificates gnupg lsb-release \
    software-properties-common ufw nginx
```

3.2 Python 3.12 + venv

```
sudo apt install -y python3 python3-venv python3-pip
python3 --version    # should print 3.12.x on Ubuntu 26
```

3.3 Node.js 20 LTS + Yarn

```
curl -fsSL https://deb.nodesource.com/setup_20.x | sudo -E bash -
sudo apt install -y nodejs
sudo npm install -g yarn
node -v && yarn -v
```

3.4 MongoDB 7

```
curl -fsSL https://www.mongodb.org/static/pgp/server-7.0.asc | \
    sudo gpg -o /usr/share/keyrings/mongodb-server-7.0.gpg --dearmor

echo "deb [ arch=amd64,arm64 signed-by=/usr/share/keyrings/mongodb-server-7.0.gpg ] \
https://repo.mongodb.org/apt/ubuntu jammy/mongodb-org/7.0 multiverse" | \
    sudo tee /etc/apt/sources.list.d/mongodb-org-7.0.list

sudo apt update
sudo apt install -y mongodb-org
sudo systemctl enable --now mongod
sudo systemctl status mongod --no-pager
```

MongoDB binds to 127.0.0.1 by default — do **not** change this. Never expose MongoDB to the public internet.

4. Get the Application Code

Choose one of the following methods to put the project on the server:

Option A — Clone from your Git repository (recommended)

```
sudo mkdir -p /var/www
sudo chown $USER:$USER /var/www
cd /var/www
git clone https://github.com/<your-org>/maria-glass-plywood.git app
cd app
```

Option B — Upload via SCP from your laptop

```
# Run on your laptop, not the server
scp -i ~/maria-key.pem -r ./app ubuntu@<ELASTIC_IP>:/tmp/app
ssh -i ~/maria-key.pem ubuntu@<ELASTIC_IP> "sudo mv /tmp/app /var/www/app && sudo chown -R $USER:$USER
```

After this step, the project tree on the server should be: `/var/www/app/backend` and `/var/www/app/frontend`.

5. Configure the Backend (FastAPI)

5.1 Create a virtual environment and install requirements

```
cd /var/www/app/backend
python3 -m venv venv
source venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
```

5.2 Create the production .env

```
sudo nano /var/www/app/backend/.env
```

Paste (replace the placeholders):

```
MONGO_URL="mongodb://127.0.0.1:27017"
DB_NAME="maria_glass_plywood"
CORS_ORIGINS="https://mariaglassplywood.com,https://www.mariaglassplywood.com"

JWT_SECRET="<run: openssl rand -hex 32>"
ADMIN_EMAIL="admin@mariaglass.com"
ADMIN_PASSWORD="<strong-password-here>"

COMPANY_EMAIL="sparkcurvsalesmail@gmail.com"
COMPANY_NAME="Maria Glass & Plywood"

SMTP_HOST="smtp.gmail.com"
SMTP_PORT="587"
SMTP_USERNAME="sparkcurvsalesmail@gmail.com"
SMTP_APP_PASSWORD="<16-char Gmail app password, no spaces>"
```

Generate a **fresh** `JWT_SECRET` with `openssl rand -hex 32` on the server. **Never** reuse the development secret in production.

5.3 Smoke test

```
cd /var/www/app/backend
source venv/bin/activate
uvicorn server:app --host 127.0.0.1 --port 8001
# In another terminal:
```

```
curl http://127.0.0.1:8001/api/  
# {"message":"Maria Glass & Plywood API"}  
# Press Ctrl+C to stop
```

6. Run the Backend as a systemd Service

```
sudo nano /etc/systemd/system/maria-backend.service
```

Paste:

```
[Unit]  
Description=Maria Glass & Plywood FastAPI backend  
After=network.target mongod.service  
Requires=mongod.service  
  
[Service]  
Type=simple  
User=ubuntu  
Group=ubuntu  
WorkingDirectory=/var/www/app/backend  
EnvironmentFile=/var/www/app/backend/.env  
ExecStart=/var/www/app/backend/venv/bin/uvicorn server:app \  
    --host 127.0.0.1 --port 8001 --workers 2 --proxy-headers  
Restart=always  
RestartSec=5  
StandardOutput=append:/var/log/maria-backend.log  
StandardError=append:/var/log/maria-backend.log  
  
[Install]  
WantedBy=multi-user.target  
  
sudo touch /var/log/maria-backend.log && sudo chown ubuntu:ubuntu /var/log/maria-backend.log  
sudo systemctl daemon-reload  
sudo systemctl enable --now maria-backend  
sudo systemctl status maria-backend --no-pager  
curl http://127.0.0.1:8001/api/
```

7. Build & Configure the Frontend

7.1 Production .env

```
sudo nano /var/www/app/frontend/.env

REACT_APP_BACKEND_URL=https://mariaglassplywood.com
WDS_SOCKET_PORT=443
```

7.2 Build the static bundle

```
cd /var/www/app/frontend
yarn install --frozen-lockfile
yarn build
# Output goes to /var/www/app/frontend/build
ls build/
```

8. Nginx Reverse Proxy

```
sudo nano /etc/nginx/sites-available/mariaglassplywood.com
```

Paste:

```
server {
    listen 80;
    listen [::]:80;
    server_name mariaglassplywood.com www.mariaglassplywood.com;

    root /var/www/app/frontend/build;
    index index.html;

    client_max_body_size 25M;          # allow construction photos
    gzip on;
    gzip_types text/plain text/css application/javascript application/json image/svg+xml;

    # Static assets
    location / {
        try_files $uri /index.html;
    }

    # Backend API
    location /api/ {
        proxy_pass http://127.0.0.1:8001;
        proxy_http_version 1.1;
        proxy_set_header Host $host;
        proxy_set_header X-Real-IP $remote_addr;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_read_timeout 60s;
    }

    # Long cache for hashed assets
    location /static/ {
        expires 1y;
        add_header Cache-Control "public, immutable";
    }
}

sudo ln -s /etc/nginx/sites-available/mariaglassplywood.com /etc/nginx/sites-enabled/
sudo rm -f /etc/nginx/sites-enabled/default
sudo nginx -t
sudo systemctl reload nginx
```

Open <http://mariaglassplywood.com> in a browser — the site should load.

9. Enable HTTPS with Let's Encrypt

```
sudo snap install --classic certbot
sudo ln -sf /snap/bin/certbot /usr/bin/certbot

sudo certbot --nginx \
  -d mariaglassplywood.com -d www.mariaglassplywood.com \
  --agree-tos -m admin@mariaglassplywood.com --no-eff-email --redirect

# Test auto-renewal
sudo certbot renew --dry-run
```

Certbot automatically updates the Nginx config to listen on 443 and redirect HTTP → HTTPS, and installs a systemd timer to renew certificates every 60 days.

10. Firewall (UFW)

```
sudo ufw allow OpenSSH
sudo ufw allow 'Nginx Full'
sudo ufw --force enable
sudo ufw status verbose
```

11. First-Login Sanity Check

Open <https://mariaglassplywood.com> — verify:

- Homepage loads with logo, hero, navigation, footer with WhatsApp button.
- <https://mariaglassplywood.com/portal> — log in with the admin credentials from your .env. You should land on /admin.
- In Admin → Field Workers, create a test worker. Log out, log back in as that worker, submit a test report with a photo. Confirm an email arrives at sparkcurvsalesmail@gmail.com within ~30s.
- Admin → Submissions → **Export Excel** and **Export PDF** both download files.

12. Useful Operational Commands

Logs

```
# Backend
sudo journalctl -u maria-backend -f
sudo tail -f /var/log/maria-backend.log

# Nginx
sudo tail -f /var/log/nginx/access.log
sudo tail -f /var/log/nginx/error.log

# MongoDB
sudo journalctl -u mongod -f
```

Restart services

```
sudo systemctl restart maria-backend
sudo systemctl reload nginx
sudo systemctl restart mongod
```

Deploy new code (after a Git push)

```
cd /var/www/app
git pull origin main

# Backend
cd backend
source venv/bin/activate
pip install -r requirements.txt
sudo systemctl restart maria-backend

# Frontend
cd ../frontend
yarn install --frozen-lockfile
yarn build
sudo systemctl reload nginx
```

Database backup (cron-friendly)

```
mkdir -p /var/backups/mongo
mongodump --uri="mongodb://127.0.0.1:27017" --db=maria_glass_plywood \
  --out=/var/backups/mongo/$(date +%F)
# Schedule daily: sudo crontab -e
# 30 2 * * * /usr/bin/mongodump --uri=mongodb://127.0.0.1:27017 --db=maria_glass_plywood --out=/var/bac
```

13. Hardening Checklist

- Disable SSH password auth — keep key-only in /etc/ssh/sshd_config (PasswordAuthentication no).

- Install fail2ban to throttle brute-force SSH attempts: `sudo apt install -y fail2ban`.
- Restrict the EC2 Security Group's SSH rule (port 22) to your IP only, not 0.0.0.0/0.
- Enable AWS CloudWatch alarms on CPU and disk usage.
- Rotate the JWT secret and admin password every 90 days.
- Schedule weekly `sudo unattended-upgrade --dry-run` to confirm security patches are applying.

14. Common Issues

Backend won't start

```
sudo systemctl status maria-backend
sudo journalctl -u maria-backend -n 100 --no-pager
# Most common cause: missing .env value or wrong file path
```

502 Bad Gateway from Nginx

Backend is not listening on 127.0.0.1:8001. Verify `curl http://127.0.0.1:8001/api/` on the server.

Field-report emails not arriving

1. Verify Gmail App Password is correct and 2-factor auth is enabled on the Gmail account.
2. Check `/var/log/maria-backend.log` for Email sending failed entries.
3. From the server, run `openssl s_client -starttls smtp -connect smtp.gmail.com:587` — must succeed.

Photo upload returns 413

Increase `client_max_body_size` in the Nginx config (currently 25M) and reload Nginx.

Appendix — Quick reference

Backend code path: `/var/www/app/backend`

Backend env: `/var/www/app/backend/.env`

Frontend build: `/var/www/app/frontend/build`

Frontend env: `/var/www/app/frontend/.env`

Nginx config: `/etc/nginx/sites-available/mariaglassplywood.com`

systemd unit: `/etc/systemd/system/maria-backend.service`

Backend log: `/var/log/maria-backend.log`

MongoDB data: `/var/lib/mongodb`

SSL cert: `/etc/letsencrypt/live/mariaglassplywood.com/`

Document version 1.0 · For Maria Glass & Plywood, Nagercoil · Built with FastAPI + React + MongoDB.